

Day 43 - Data Engineering for AI Products

Submission Ready Document

Data Ingestion Pipeline

```
import re, time
from langchain_community.document_loaders import DirectoryLoader, TextLoader
from langchain.text_splitter import RecursiveCharacterTextSplitter
from langchain_openai import OpenAIEmbeddings
from langchain_community.vectorstores import FAISS

loader=DirectoryLoader('data',glob='**/*.txt',loader_cls=TextLoader)
documents=loader.load()
splitter=RecursiveCharacterTextSplitter(chunk_size=500,chunk_overlap=100)
chunks=splitter.split_documents(documents)
embeddings=OpenAIEmbeddings()
vectorstore=FAISS.from_documents(chunks,embeddings)
vectorstore.save_local('faiss_index')
```

Incremental Update Function

```
def incremental_update(new_docs):
    embeddings=OpenAIEmbeddings()
    db=FAISS.load_local('faiss_index',embeddings,allow_dangerous_deserialization=True)
    db.add_texts(new_docs)
    db.save_local('faiss_index')
```

Retrieval Review

10 queries tested. Query #9 (RAG vs Fine-Tuning) flagged as retrieval failure.

Cost Documentation

Initial Cost: \$0.04
Monthly Cost: \$0.04
10x Scale Cost: \$0.40